Garmin Heart Rate Balance Dashboard

Matthew Block - CSIS 44-613 Data Visualization
25 February 2026

1. Introduction

The goals for this report are primarily based on the recommendations by Matt Fitzgerald and David Warden in the book 80/20 Triathlon: Discover the Breakthrough Elite-Training Formula for Ultimate Fitness and Performance–at All Levels. The book argues for an 80%/20% balance between low-intensity and high-intensity work for endurance training. While the book suggests training plans that help plan an appropriate balance, I find that I am often unable to execute the training plan perfectly due to schedule considerations, weather, or other factors.

The overall theme goal of this dashboard is to track the balance of fitness training actually executed to inform the training plan moving forward and avoid exceeding 20% of training time in the high-intensity range.

All charts can be seen on the interactive dashboard with tooltips at https://public.tableau.com/app/profile/matthew3498/viz/Block-Garmin-Heart-Rate-Balance-Dashboard/TotalVolume?publish=yes

2.Data Description

I pulled my recorded activity data from my Garmin profile. Unfortunately the export I used in previous projects didn't have the heart rate zone data I needed. Apparently you have to pull that level of detail via API. I found a GitHub repository by daviderubio that establishes a regular API pull to Google Sheets from Garmin. I took that and altered it to include the Heart Rate Zone data, VO2 max, and other data fields I was interested in. I set this up for a daily pull via GitHub actions so that I can build a dashboard that is updated daily and can inform my fitness planning dynamically each day.

3.Data Cleaning Strategies

Cleaning was not required, but some transformations were helpful:

- Activity types were compiled into groups of like activities
- Pivoting was applied to the heart rate zones in order for them to be able to be summed and charted as needed for the analysis.
- Date data was originally imported as text and was converted to date types.

4. Clean Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Activity ID	Date	Activity Name	Type	Distance (km)	Duration (min)	Pace (min/km)	Avg HR	Max HR	Calories	Avg Cadence	Elevation Gain (r	Z1 (min)
2	21965542196	2026-02-23	Honolulu - W01D1-Easy Run	running	3.46	41.47	11.99	96	119	249	98	49.4	9.2
3	21954265312	2026-02-22	Honolulu Walking	walking	1.99	21.54	10.84	128	147	163	87.796875	33	7.32
4	21943859743	2026-02-21	Coherence	breathwork	0	15.57	0	94	107	0	0	0	0.91
5	21943770023	2026-02-21	Honolulu - Base	running	5.11	44.79	8.77	150	164	402	139.703125	56.8	1.35
6	21921828818	2026-02-19	Honolulu - Base	running	4.05	33.09	8.18	146	160	328	136.171875	50.7	1.52
7	21892130373	2026-02-16	Rucking the dog	walking	1.98	26.06	13.17	114	151	167	77.296875	31.7	12.23
8	21889719484	2026-02-16	Honolulu - Hill sprints 10*15s	running	3.17	28.72	9.06	144	177	278	113.828125	64.7	2.18
9	21880043348	2026-02-15	Honolulu - Tempo	running	8.13	60.74	7.47	158	176	662	144.640625	88.4	0.97
10	21860487689	2026-02-13	Honolulu - Long Run	running	7.92	70.6	8.92	151	168	636	129.9375	95.9	0.65
11	21829475018	2026-02-10	Indoor Cycling	indoor_cycling	9.66	28.54	2.96	134	168	251	0	0	2.27
12	21808906486	2026-02-08	Honolulu - Base	running	6.23	53	8.5	148	163	504	141.28125	75.8	1.51
13	21755172774	2026-02-03	Honolulu Multisport	multi_sport	4.45	80.45	18.07	0	140	544	0	75	0
14	21734813204	2026-02-01	Honolulu - Base	running	4.52	35.19	7.79	144	159	367	144.015625	50.8	1.48
15	21724351150	2026-01-31	Honolulu Walking	walking	1.58	20.39	12.94	111	130	126	76.734375	23.1	15.27
16	21694554359	2026-01-28	Honolulu - Hill sprints 10*15s	running	3.68	31.2	8.48	144	177	320	135.390625	65.7	3.67
17	21683561200	2026-01-27	Honolulu - Base	running	6.4	49.14	7.67	149	164	528	146.609375	55.1	1.54
18	21663871750	2026-01-25	Honolulu - Base	running	3.66	28.95	7.9	142	157	292	145.34375	39.2	1.37
19	21652909277	2026-01-24	Magic Island Biathlon (Run-Swim)	multi_sport	5.72	45.79	8	0	184	599	0	8.7	0
20	21613974413	2026-01-20	Ka'ena Ruck	biking	19.21	224.41	21.97	101	134	910	51.734375	117.6	91.88

Full dataset viewable at
https://docs.google.com/spreadsheets/d/1WW3weSxRh7gB12xnFuTpwoUAY5x1EodGUrD93fCIhB8/

5.Visualization Tool

For this project, I chose to perform the work in Tableau because I find its interface and workflow smoother and more modern (although not necessarily more intuitive). By using Tableau I was able to get more practice with this tool that is newer to me and test out additional features not covered in the course, such as connecting to Google Sheets for a data source and the pivot function (which I found quite different from Excel's pivot tables).

I also intend to migrate this dashboard to Google Looker Studio for ongoing use. The interface for Looker is more similar to Tableau so this preparation work will be more applicable.

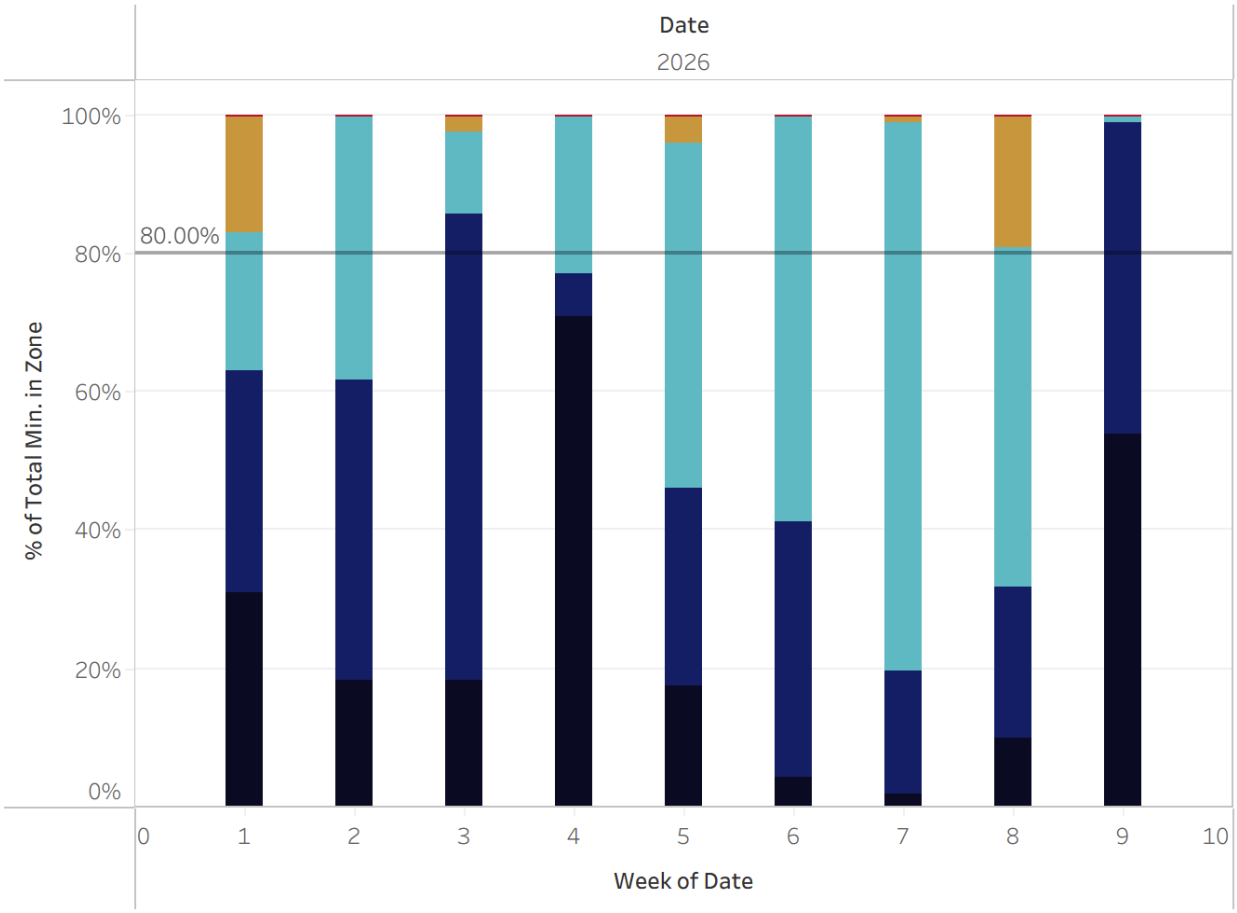
- The python scripting to pull the data from Garmin to Google Sheets via GitHub Actions is in the repository at https://github.com/matthewpblock/garmin-gsheets-sync
- The Google Sheet that is updated daily is at https://docs.google.com/spreadsheets/d/1WW3weSxRh7gB12xnFuTpwoUAY5x1EodGUrD93fCIhB8/
- The dashboard on Tableau Public can be accessed at https://public.tableau.com/app/profile/matthew3498/viz/Block-Garmin-Heart-Rate-Balance-Dashboard/TotalVolume?publish=yes

6. Visualizations & Stories

The core efforts in triathlon training are the swim, bike, and run. Ideally, each of these efforts maintains the 80/20 ratio inside of its own effort type:

Goal/Question 1: What percentage of **running** time was spent in each heart rate zone? **(Target 80% Zone 1-2)**

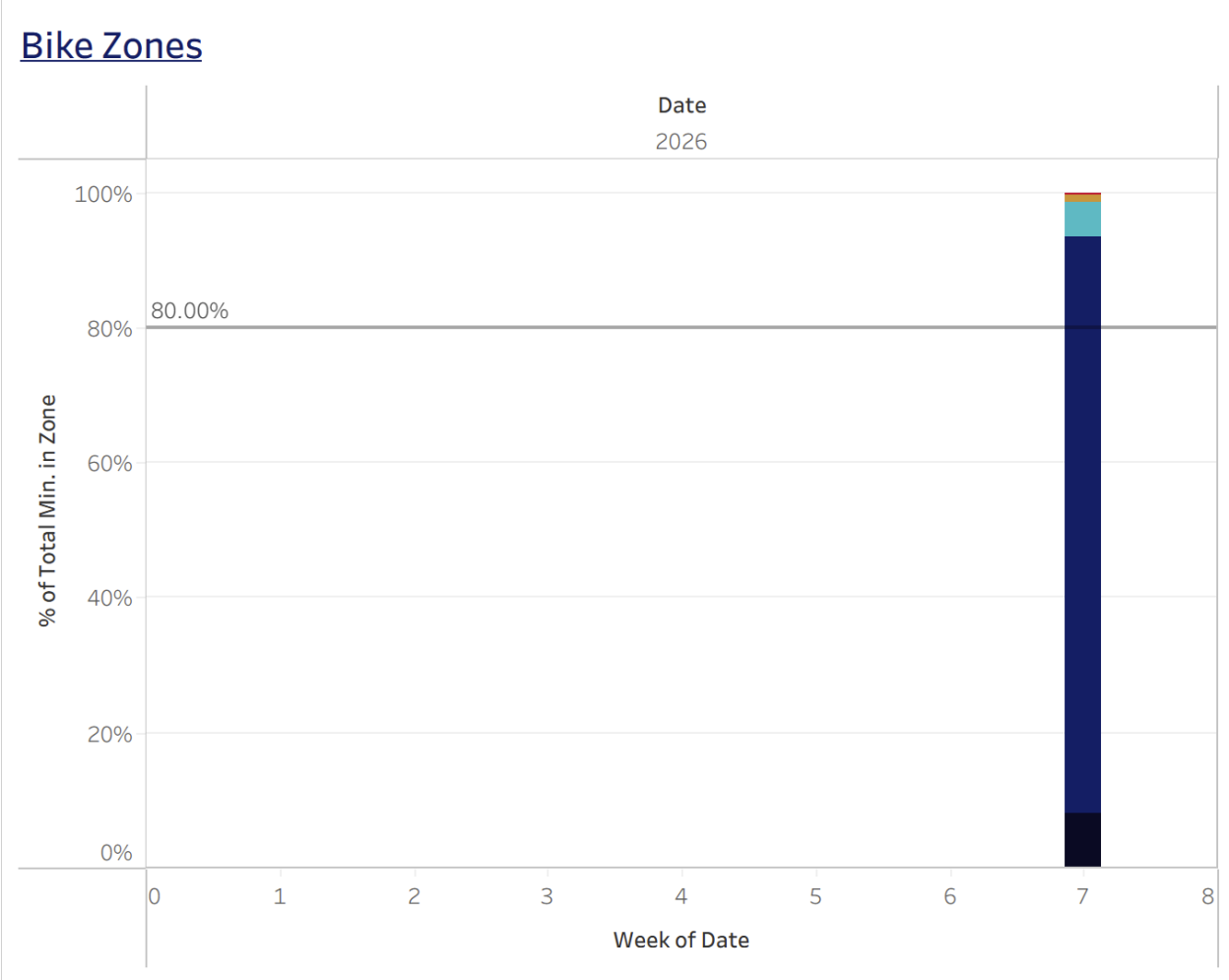
Run Zones



Based on the 80/20 Triathlon recommendations, the bottom two zones constitute the low-intensity target. Based on this recommendation, I only met the target balance of at least 80% low-intensity effort in week 3 and week 9 (currently ongoing). The book suggests that zone 3 (the coral color) should largely be avoided. Clearly my efforts have not been meeting the recommended polarization most weeks. For week 9, I have a good starting base of low-intensity effort and have room to add some interval work before the week ends.

Recommendation: Perform the majority of runs at a slower pace to avoid breaking into zone 3. Once a solid base of low-intensity work has been achieved for a week, add in an appropriate amount of high-intensity interval training.

Goal/Question 2: What percentage of **biking** time was spent in each heart rate zone? **(Target 80% Zone 1-2)**

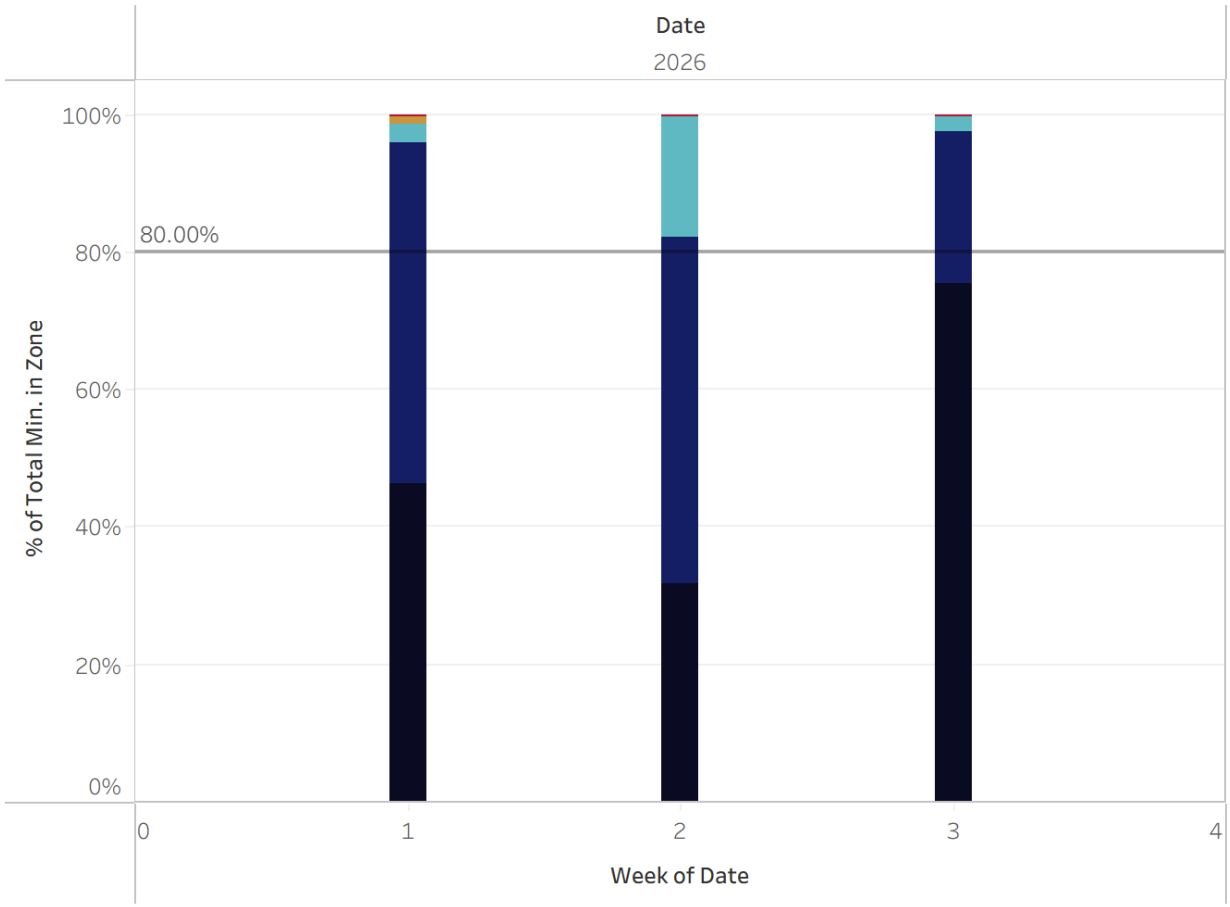


The only week with biking activity was week 7. Biking balance was not too aggressive. In fact, there was even more room for high-intensity activity in the week's workouts. Adding some more bursts of high-intensity interval activity at the end of bike workouts could help me gain fitness faster moving forward.

Recommendation: Bike consistently every week. Reserve time at the end of bike sessions for high-intensity interval training.

Goal/Question 3: What percentage of **swimming** time was spent in each heart rate zone? **(Target 80% Zone 1-2)**

Swim Zones

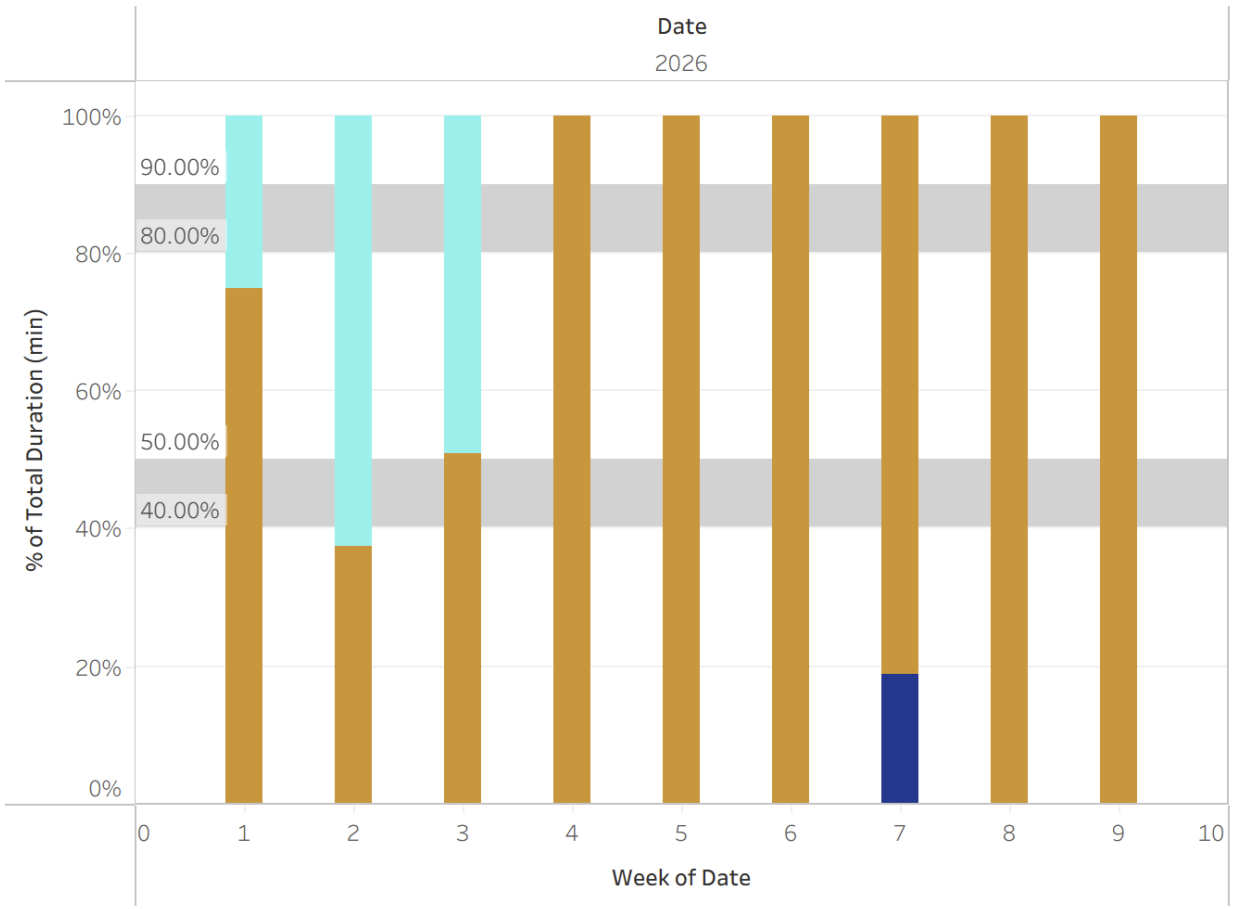


I had a good start to swimming for the year before falling off in week 4. Swim efforts all fit well within the recommendation of at least 80% low-intensity effort. The week I tried to add more high-intensity my heart rate only reached moderate-intensity levels. I may need to explore new ways to get high-intensity swim efforts in. Many of my swims are during a lunch break, so performing high-intensity swim training then returning to work may prove difficult.

Recommendation: Attempt to include some high-intensity interval training at the end of each swim session, but aim for 10% or less of total time. Plan an evening or weekend swim each week specifically to focus on high-intensity swimming.

Goal/Question 4: What percentage of training time was spent in each sport **(Target 40-50% bike, 10-20+% swim)**

Tri Balance

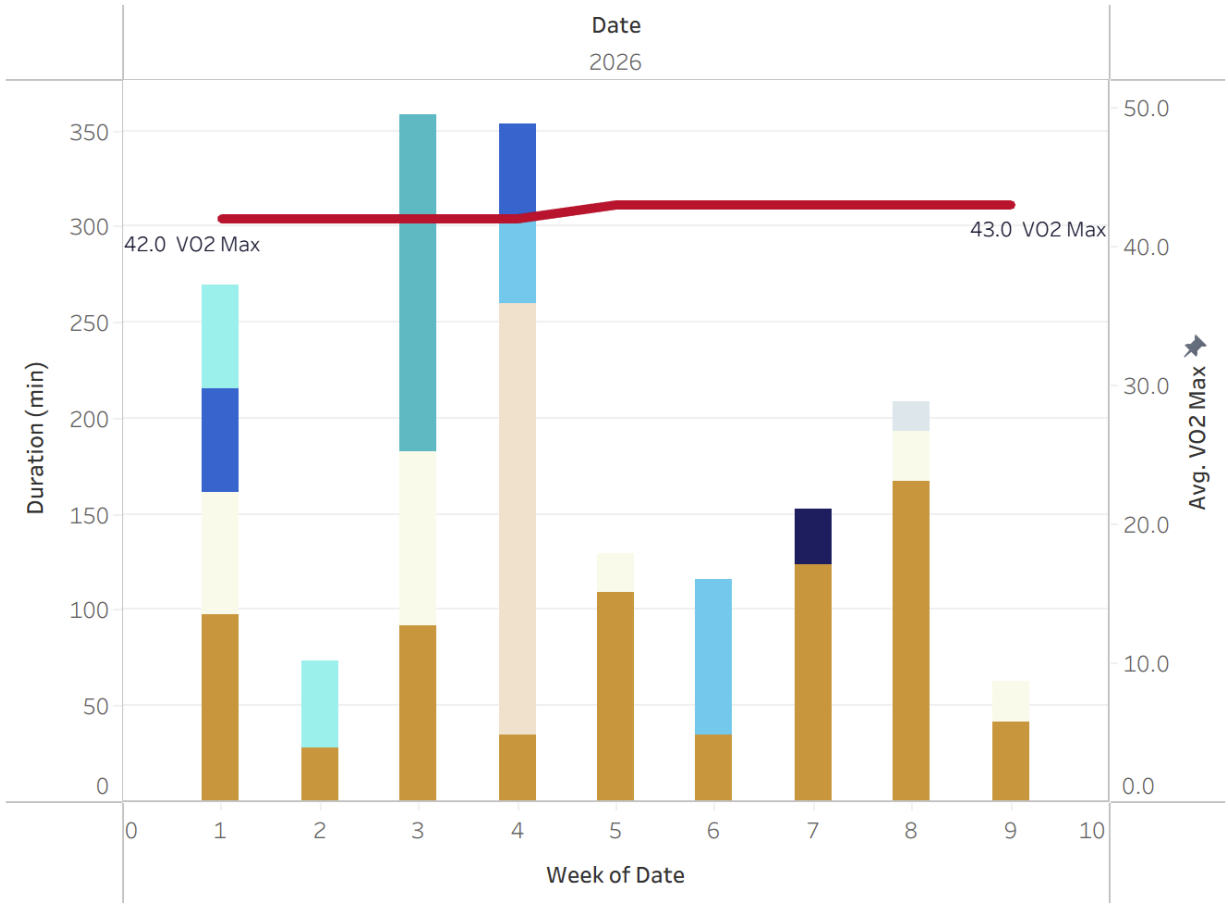


While the balance of triathlon training activities is completely over-weighted toward running, this isn't entirely an accident. I've been training for a 10k race on March 1 and have had a run focus, using the other activities to augment my run training. Moving forward, the intent is to focus training on a more triathlon-appropriate balance. Dramatically increasing bike activity and being consistent with swim activity will be key to this shift.

Recommendation: Reference this dashboard regularly to ensure maintaining the proper balance. Workouts late in the week can be flexed among sports to compensate for early-week activities that weren't executed as planned.

Goal/Question 5: How much did total training time increase over prior week? **(Target 5-10%)**

Total Volume

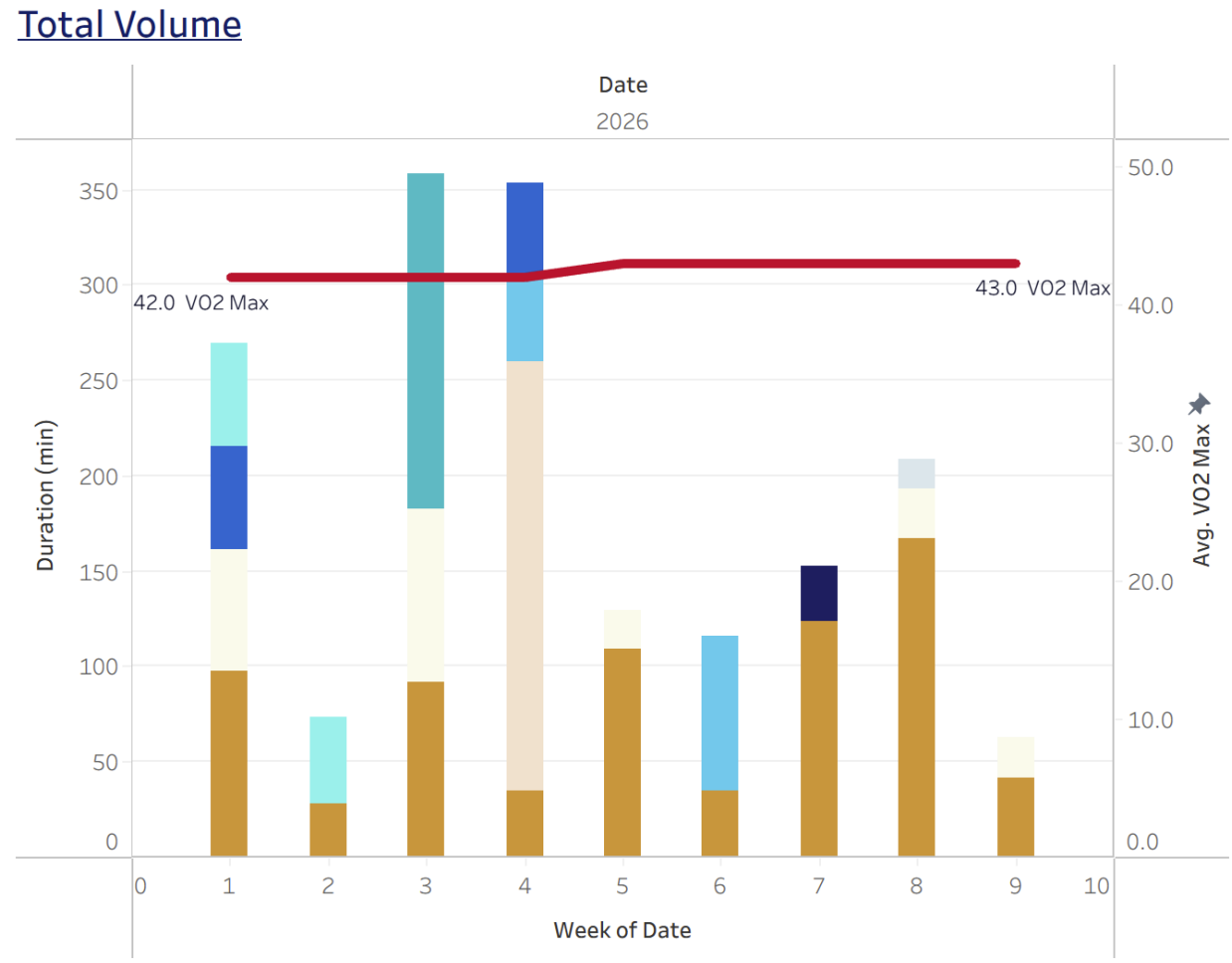


Increasing volume too quickly can lead to over-training stress. Looking at week 9 (currently in progress), I likely do not want to exceed approximately 250 minutes of training activity for the week to maintain the steady growth from weeks 5-8.

Recommendation: Work to exceed 200 minutes this week. While 250 minutes is likely a good target, I almost certainly should not exceed 350 minutes (the highest volume weeks on the chart).

Goal/Question 6: How did training activity impact VO2 max?

For Goal 6, we will look at the same chart again since I joined the data for Goals 5 & 6 into the same dual-axis chart:



Higher VO2 max scores indicate higher fitness. The increase from 42 to 43 seems to correspond to the highest volume weeks (and perhaps the lower volume recovery week just after).

Recommendation: Work up to 350 minute weeks again and look for patterns in future increases in VO2 max. Any drops in VO2 max should cue a lower-volume recovery week.

7. Conclusions:

Without this dashboard to reference, many of my activities were not properly balanced. Running was particularly mis-aligned with far too much activity at moderate-intensity levels. Even with a balanced plan, adding a daily review of this dashboard will help me maintain a better balance by exposing each day how a missed workout or unintentionally aggressive run affects the overall balance across the week.

Recommendations for further development:

- Mapping other health indicators such as Heart Rate Variability or Resting Heart Rate
- Evaluate other potential measures of training volume such as Calories burned or Training Load
- Develop a method of tracking or evaluating multi-sport “brick” workouts